

Humidification Experiment: Sample Calculations

The following example is given to provide guidance on your calculations for this experiment. The following data is used:

Initial dry air properties (no water flow): Temperature = 21.3 °C Relative Humidity = 9.2%

Humidification Experimental Data:

Air flow rate: 9 SCFM

Air temperature: 55 °C

Inlet air pressure: 28 psig (42.7 psia)

Thermocouple 1: 31.8 °C

Thermocouple 3: 31.0 °C

Thermocouple 4: 29.8 °C

Thermocouple 5: 29.7 °C

Thermocouple 6: 30.9 °C

Relative humidity at outlet: 29.8% at a temperature of 43.8 °C

Calculations

Mass Transfer

$$\text{Formula for mass transfer coefficient } (k_y): \ln \left(\frac{Y_i' - Y_1'}{Y_i' - Y_2'} \right) = \frac{k_y Pz}{G_s} \quad \text{Equation 1}$$

Y_i', Y_1', Y_2' are the mass fractions of water with respect to air at the interface, inlet, and outlet, respectively. Y_i' can be calculated at the average temperature of thermocouples 3-5, which gives a value of $T_i = 30.2$ °C. The three Y' variables can be calculated using an online psychrometric chart or a humidity calculator for better accuracy than manually looking at the psychrometric graph. See the two references given. Using reference 1 (humidity calculator), Y_i' was calculated at the saturation temperature of 30.2 °C, Y_1' was calculated at 55 °C with a relative humidity of 9.2%, and Y_2' was calculated at 43.8 °C with a relative humidity of 29.8%:

$$Y_i' = 0.0268 \frac{\text{kg Water}}{\text{kg Air}}$$

$$Y_1' = 0.0089 \frac{\text{kg Water}}{\text{kg Air}}$$

$$Y_2' = 0.0167 \frac{\text{kg Water}}{\text{kg Air}}$$

$$\text{Solving for } k_y: k_y = \frac{G_s \cdot \ln \left(\frac{Y_i' - Y_1'}{Y_i' - Y_2'} \right)}{Pz} \quad \text{Equation 2}$$

G_s is the mass flow rate of air. Since the air flow rate was measured in SCFM, it has to be converted to units of kg/s. Another online calculator can be for this, since it is a function of the pressure, temperature, and humidity of the air; see reference [3].

$$k_Y = \frac{0.00456 \frac{\text{kg}}{\text{s}} \cdot \ln\left(\frac{0.0268 - 0.0089}{0.0268 - 0.0167}\right)}{0.1596\text{m} \cdot 1.2192\text{m}} = 0.0134 \frac{\text{kg}}{\text{s}}$$

The number of transfer units, or NTU, is simply the left-hand side of equation 1.

$$\text{NTU} = \ln\left(\frac{Y_i' - Y_1'}{Y_i' - Y_2'}\right) = 0.57 \quad \text{Equation 3}$$

The height of transfer unit, or HTU, is the height of the column divided by NTU:

$$\text{HTU} = \frac{L}{\ln\left(\frac{Y_i' - Y_1'}{Y_i' - Y_2'}\right)} = 2.14 \text{ m} \quad \text{Equation 4}$$

Heat Transfer

Formula for the heat transfer coefficient (h): $\ln\left(\frac{T_i - T_1}{T_i - T_2}\right) = \frac{hPz}{G_s C_s}$ Equation 5

Equation 5 can be solved for h: $h = \frac{G_s C_s}{Pz} \ln\left(\frac{T_i - T_1}{T_i - T_2}\right)$ Equation 6

$$C_s = C_{Air} + Y_i' C_{Water} \quad \text{Equation 7}$$

$$C_s = 1005 \frac{\text{J}}{\text{kg} \cdot \text{K}} + 0.0267 \cdot 1864 \frac{\text{J}}{\text{kg} \cdot \text{K}} = 1055 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$h = \frac{0.00456 \frac{\text{kg}}{\text{s}} \cdot 1055 \frac{\text{J}}{\text{kg} \cdot \text{K}}}{0.156\text{m} \cdot 1.22\text{m}} \cdot \ln\left(\frac{30.2 - 55}{30.2 - 43.8}\right) = 15.2 \frac{\text{J}}{\text{K} \cdot \text{m}^2 \cdot \text{s}}$$

References:

1. Humidity calculator (Quadco Engineering): Relative humidity to mass fraction H₂O <https://www.quadco.engineering/en/know-how/cfd-calculate-water-fraction-humid-air.htm>
2. Psychrometric Chart Calculator. Adicot Inc. <https://www.adicotengineering.com/psychrometric-chart-calculator>
3. Air - SCFM versus ACFM and ICFM. Engineering ToolBox. www.engineeringtoolbox.com/scfm-acfm-icfm-d_1012.html.